LESSON PLAN · SCRATCH JUNIOR (K–2) ADVANCED: GAME BUILDERS

Lesson 6: Project: Gem Collector Game

Module 2 | 35 minutes | Project: score and paths | KG–Grade 2 · ages 6–7

Objective: Plan and build a game where a character collects treasure, the score goes up and the game says when you win.

Success criteria

- My hero collects treasure.
- Treasure disappears.
- My game says when you win.

Key vocabulary

- **collect**: Pick up.
- **win**: Finish the game's goal.
- **bump**: When two characters touch.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (35 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Coin hunt (5 min): collect paper coins and count the score.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
10 min	**Build**	Build-along steps, then Coin run. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; My Gem Collector Game.

Unplugged warm-up

Coin hunt (5 min): collect paper coins and count the score.

Teaching the idea

Combines path planning, counter variables (puzzle) and bump events / hide (app) into a win-condition game.

Analogy: A treasure hunt with a scoreboard.

Live demo: Show treasure with “start on bump → hide”.

Questions to ask

Ask	Listen for
How does treasure disappear?	Bump → hide.
When does the hero say you win?	At the end.
How many points?	One per treasure.

Differentiation

Support
Two treasures only.

Stretch
Add a “bad” item that makes the hero shrink.

Common misconceptions

Misconception	How to fix it
Treasure hides by itself	Use bump → hide.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Game rules	Not yet	Collects one	Collects all and wins	Extra rule (bad item, timer)
Coding	Not yet	Moves	Moves, bumps and hides	Neat code with loops
Play testing	Not yet	Played with help	A friend played it	Changed something after testing

Exit ticket
What happens when the hero bumps a treasure?

Homework
Tell your family the rules of your game.

Answer key — Kitabash activities

1. Coin run (perform · block code)

One solution:



2. My Gem Collector Game (probe · project)

Answer key — worksheet

1. **Key words:** collect = Pick up. · win = Finish the game's goal. · bump = When two characters touch.
2. **My plan:** A clear plan that covers every requirement.
3. **Checklist:** All requirements met.
4. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.